Vision Transformer-Based Visual Analysis of Hybrid-Slotted DGS Compact Patch Antennas

Module 1 of the Applied Vision Transformer Framework for 5G and Emerging 6G Networks - Proof-of-Concept Prototype

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1 Research objective

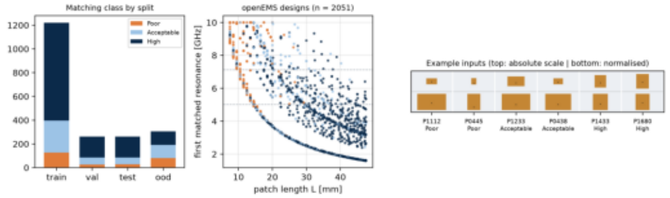
- Can standardised antenna geometry images predict simulated performance classes - and show where the model looks?
- Module 1 = image branch of the two-module framework; Module 2 (surrogate, fusion, NSGA-II) is a separate study.
- Proof of concept on public simulation data; Ansys HFSS was not available and was not run.
- Hypotheses H1-H5 committed to Git before training.

2 Data & preprocessing

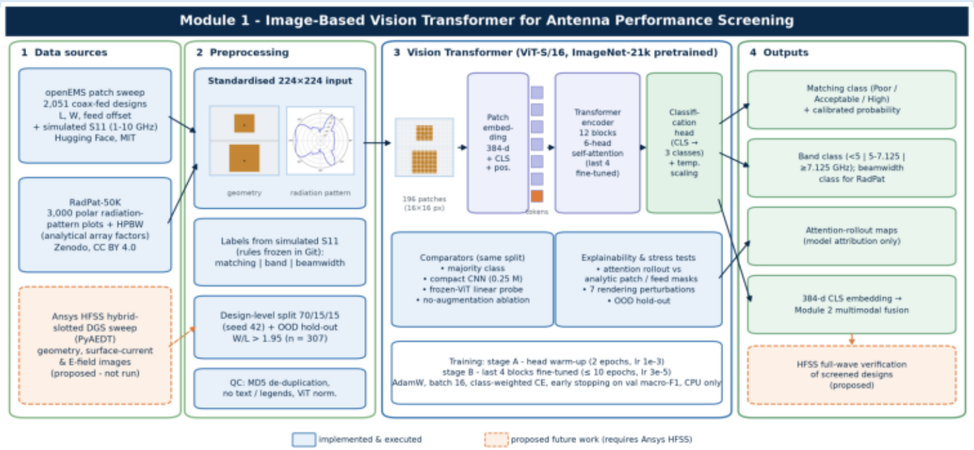
- openEMS patch sweep (Hugging Face, MIT): 2,051 designs -> one 224x224 two-panel image each
- RadPat-50K (Zenodo, CC BY 4.0): 3,000 polar radiation-pattern plots -> beamwidth class
- Locked design-level split 70/15/15 (seed 42): 1,220 / 262 / 262 + OOD hold-out W/L > 1.95 (n = 307)
- No text / legends, MD5 de-duplication, training-only mirror, ±3° and 2% shift

Matching class: Poor > -10 dB | Acceptable -20 to -10 dB | High <= -20 dB*

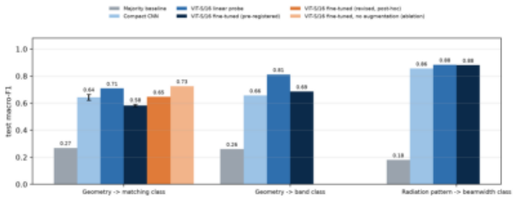
**-20 dB boundary is a project assumption awaiting engineering confirmation*

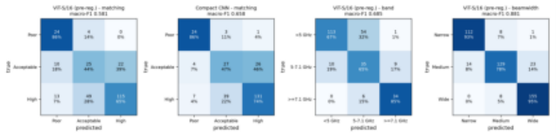


3 Module 1 block diagram (implemented vs proposed)



4 Results on the locked test split (executed)

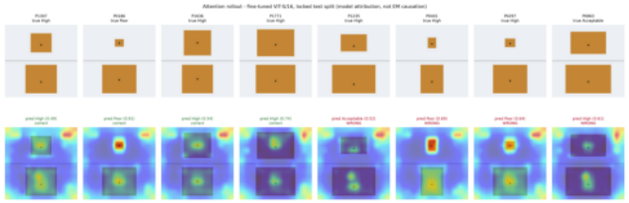




Model (geometry -> matching class)	Macro-F1	Bal. acc.	Other tasks
Majority baseline	0.269	0.333	
Compact CNN (3 seeds)	0.643 ± 0.022	0.669	Band class ViT 0.685 probe 0.811
ViT-S/16 linear probe	0.709	0.729	
ViT-S/16 fine-tuned (3 seeds)	**0.582 ± 0.008**	**0.649**	
ViT fine-tuned - OOD hold-out	0.679 ± 0.027	0.664	RadPat beamwidth ViT 0.881
ViT, no augmentation (ablation*)	0.725	0.737	

**exploratory, 1 seed, not pre-registered; underfitting traced to augmentation blurring the 1-2 px feed offset*

5 Explainability (attention rollout)



Median enrichment vs area share: feed halo ×1.41, patch ×1.03

Attribution only - not proof of electromagnetic causation

6 Robustness & key insight

- Blur (1 px) -> 0.526, grey copper -> 0.521 (clean 0.581); contrast and JPEG barely matter
- OOD 0.679 > test: class balance differs, so the shift is confounded - not better generalisation
- Without augmentation the same ViT reaches 0.725: augmentations must preserve fine geometry

7 Hypotheses & conclusion

H1	ViT > majority baseline (both patch tasks)	**Supported**
H2	ViT ≥ compact CNN (3-seed mean, matching class)	**Not supported**
H3	Partial fine-tune > frozen linear probe	**Not supported**
H4	Attention enriched on patch and feed (median > 1)	**Supported**
H5	Lower macro-F1 on OOD (W/L > 1.95)	**Not supported**

- Working, reproducible image pipeline - ready for HFSS data
- Next: >= 1,000 HFSS hybrid-slotted DGS designs (PyAEDT)

Explainable image-based screening of compact patch antennas for 5G / 6G - simulation evidence only, not HFSS-validated

Data: LaBash et al. (2025) arXiv:2505.18188 [MIT] | Bhuma et al. (2025) RadPat-50K, doi:10.5281/zenodo.17067881 [CC BY 4.0]
ViT: Dosovitskiy et al. (2021) | Attention rollout: Abnar & Zuidema (2020) | Calibration: Guo et al. (2017) | Code: https://github.com/rathiabeth-sudo/901---module-1